

VINAYAK GUPTA

guptavinayak112@gmail.com | linkedin.com/in/vinayak112

EDUCATION

Birla Institute of Technology and Science, Pilani

MSc (Hons) in Biological Sciences | BE (Hons) in Manufacturing Engineering

2022 – Present

INTERNSHIPS

CSIR–Institute of Genomics and Integrative Biology, Delhi

Feb '26 – Present

Advisor: Professor Lipi Thukral

Molecular Dynamics Simulations for RNA-Lipid Interactions

- Performed molecular dynamics simulations of RNA molecules using GROMACS, including system preparation, solvation, ion addition, energy minimization, equilibration, and production runs.
- Investigated RNA structural stability and conformational dynamics through trajectory analysis and interpretation of simulation-derived structural parameters, such as RMSD, RMSF, radius of gyration.

Reproduction and Extension of the CplxCavity RNA Binding Site Prediction Framework

- Reconstructed a literature-derived pipeline for RNA cavity detection using structural geometry, surface residue analysis, and cavity descriptor extraction.
- Evaluating interpretable machine learning models (Logistic Regression, Random Forest, and XGBoost) for prediction of RNA small-molecule binding pockets.

PROJECTS

Antimicrobial Resistance Prediction for Whole Genome Sequences using ML

Advisor: Professor Sandhya Marathe & Professor Tanmaya Mahapatra

Aug '25 – Dec '25

- Worked on developing datasets for **antimicrobial resistance (AMR)** across diverse drug–bug pairs.
- Integrated genomic features and MIC values to improve accuracy and generalization in prediction models.

CNN-Based Model for Pneumonia Classification Using Chest CT Data

Dec '24

- Built a deep learning classification model to differentiate between healthy and pneumonia-affected lungs.
- Implemented image preprocessing pipelines using OpenCV and developed models in PyTorch for training.

Bioinformatics Course Project

Advisor: Professor Shibasish Chowdhury

Jan '25 – May '25

- Predicted the structural(introns, exons, Promoter regions) and coding regions using **GENSCAN**.
- Created phylogenetic trees of species using **MEGA11**, to find maximum parsimony.
- Pairwise gene alignment of multiple gene sequences using **BLASTn**.

Applied Biology Project Lab

Advisor: Professor Shilpi Garg

Jan '24 – May '24

- Conducted a series of experiments to culture microorganisms such as *E. coli* to learn wet lab practices.
- Conducted assays including **MICP**, **MIC**, **Beta-galactosidase activity** and **MTT cell viability**.

SKILLS & TECHNICAL PROFICIENCY

Languages Python[Fast.ai, Numpy, Pandas, Biopython, Scikit-Learn], C++, C, Git, HTML, CSS, bash

Bioinformatics: GROMACS, ChimeraX, PyMol, Transcriptomic Data Analysis, WGS analysis

Design & Simulation: AutoCAD, Fusion 360, Autodesk Maya, Arduino

COURSEWORK

Computational Biology: Biomolecular modeling, Bioinformatics, Recombinant DNA Technologies, Computer programming, Probability & Statistics, Machine Drawing, Engineering graphics

Biological Sciences: Developmental Biology, Immunology, Biological Chemistry, Genetics, Biophysics, Microbiology, Ecology & Environmental Sciences, Animal & Plant Physiology, Cell Biology